

FORMATTING INSTRUCTIONS FOR ICLR 2027

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ABSTRACT

Effective multi-agent interaction requires agents to infer their partners’ preferences and plan actions in light of those inferences. Terminal outcomes, however, reveal little about where either capability fails. We study inference and planning in a finite negotiation game with private preferences and mixed incentives. An oracle computes beliefs and action values under a fixed reference policy, allowing us to evaluate the two capabilities at the same decision. Qwen3-4B-Instruct-2507 misreads partner behavior and often chooses suboptimal actions even when given the correct inference. Replacing its inference with an oracle judgment changes its choices but does not reliably improve their value. We then construct *game slices*, contiguous segments of one or more decisions drawn from deliberately structured games, and use oracle labels to train action selection, inference, and planning conditioned on a correct inference. We compare this targeted training with complete-game self-play and evaluate transfer to meeting coordination with private calendars. On a 24-game sequential adaptation of CalBench, joint slice training retains 47 of 72 meetings, compared with 37 for both the unchanged baseline and the self-play model. These results support controlled games as a means of diagnosing interaction failures and training capabilities that transfer to more complex coordination.

1 INTRODUCTION

Large language models (LLMs) increasingly operate in multi-agent systems that distribute work and coordinate decisions through interaction (Zhu et al., 2025; Chen et al., 2026). Yet interaction can fail at several distinct points. In tasks with distributed information, agents leave partners’ private evidence unexplored (Li et al., 2026a); in distributed computation, they exchange messages without turning them into effective joint reasoning (Zhang et al., 2026b); and in negotiation, sharing the relevant information does not by itself ensure a coordinated plan (Yao et al., 2026). A broader analysis likewise identifies inter-agent misalignment as a distinct failure class (Cemri et al., 2025). These findings raise a basic question: what prevents individually capable agents from coordinating effectively?

Interactive decision theory provides a starting point. In an interactive partially observable Markov decision process (I-POMDP), an agent maintains beliefs about both the physical environment and other agents, and uses those beliefs to choose actions (Gmytrasiewicz & Doshi, 2005). Opponent-modeling work similarly links inferences about others’ goals or policies to the planning of a response (Carmel & Markovitch, 1996; Huang et al., 2024). Building on this distinction, we focus on two capabilities: *interactive-state inference*, which estimates partners’ task-relevant hidden states from the available evidence, and *interaction planning*, which uses those estimates to evaluate and select actions. We refer to them as *inference* and *planning*. They form a feedback loop: an inference informs an action, which shapes partners’ responses and the evidence available for subsequent decisions. Studies of LLM agents find that accurately predicting another agent’s state or behavior need not lead to an effective response (Riemer et al., 2025; Sobotka et al., 2026). This gap motivates evaluating both the inference itself and the decisions made from it.

Diagnosing these capabilities in realistic interactions is difficult because information, objectives, and decisions are distributed across agents and time. Coordination with private calendars, strategic coalition formation, and commerce on behalf of different principals illustrate this complexity (Zou

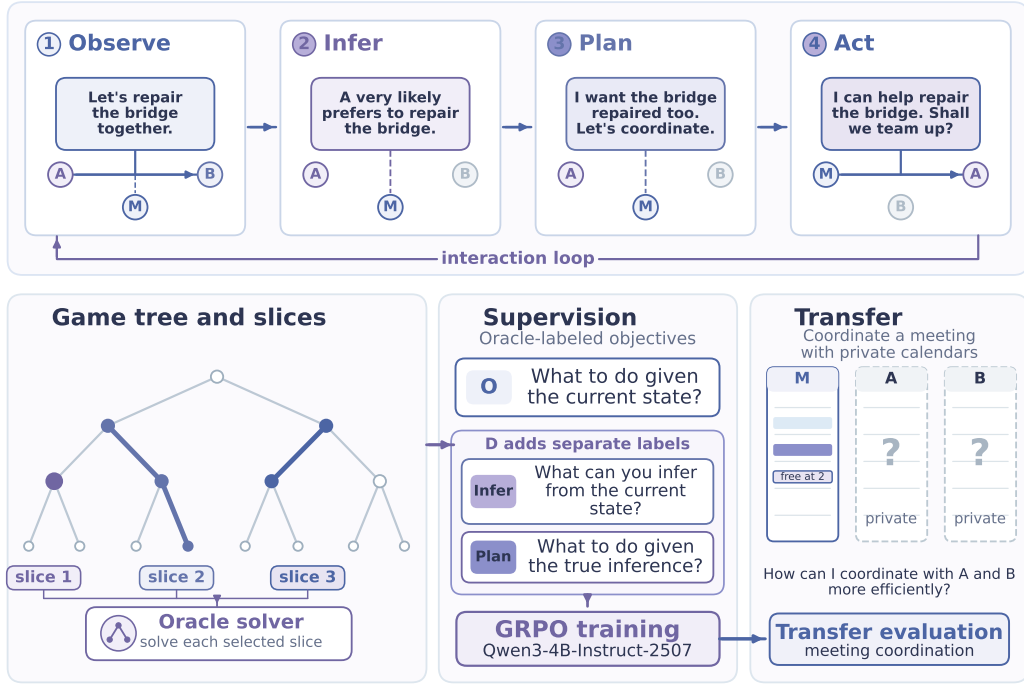


Figure 1: From interaction to targeted training and transfer. Agent M observes its partners, infers their preferences, plans a response, and acts to create new observations. We select game slices (colored nodes and paths) and solve them with an oracle to supervise action selection, inference, and planning given a correct inference. These objectives train a shared model with GRPO. We then evaluate transfer to meeting coordination, where agents use private calendars and exchanged messages to agree on meeting times.

et al., 2026; O’Neill et al., 2026; Li et al., 2026b). Games offer a controlled setting: their information structure, utilities, legal actions, and strategic consequences can be specified and verified. Recent work has used multi-agent games and self-play to train strategic or transferable reasoning in LLMs (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b). Yet a game-level outcome alone does not reveal whether an agent misread a partner or failed to act on a correct belief. In social deduction games, for example, a single trajectory combines inference, deception, and collective decision-making (Light et al., 2023). We seek a setting that retains strategic uncertainty while allowing beliefs and actions to be evaluated separately. To this end, we extend the commitment-based multi-party negotiation game of Benac et al. (2026) with private preferences and investigation actions that consume a decision opportunity. Hidden preferences create an inference problem; their alignment or conflict determines the value of possible plans.

This setting connects diagnosis, targeted training, and transfer (Figure 1). We first diagnose Qwen3-4B-Instruct-2507 by evaluating inference and planning at the same decision and replacing the model’s explicit inference with an oracle judgment (Section 3). Its choices depend on the supplied judgment, but correcting that judgment does not reliably resolve planning errors. We then construct *game slices*: contiguous segments of one or more decisions from deliberately structured small games. An oracle supplies partner-preference judgments and counterfactual action values under a fixed reference policy, providing targeted supervision for action selection, inference, and conditional planning. We compare slice training with complete-game self-play in terms of training-token efficiency, generalization to held-out game structures, and transfer to coordination with private calendars (Section 4). In our sequential CalBench evaluation, the jointly trained model retains more meetings than either the baseline or self-play, providing evidence that strategic behaviors learned in small games can transfer to more complex interactions.

2 MOTIVATION AND METHODS

2.1 FROM INTERACTIVE UNCERTAINTY TO TWO CAPABILITIES

Consider three coding agents working on the same Python backend. In an Anthropic experiment motivated by behavior observed in deployment, each was instructed to migrate it to a different language. The agents mistook the resulting interference for deliberate obstruction and escalated to repeatedly killing competing processes and, in some runs, revoking one another’s access (Anthropic, 2026). They depended on a usable shared backend, yet pursued incompatible local objectives without knowing the others’ instructions. Such coordination is naturally an interactive partially observable decision problem. For agent i , an I-POMDP is $\mathcal{I}_i = \langle IS_i, A, T_i, \Omega_i, O_i, R_i \rangle$, $IS_i = S \times \prod_{j \neq i} M_j$ where S is the physical state, M_j models agent j , A is the joint action space, and T_i, O_i, R_i specify transition, observation, and reward (Gmytrasiewicz & Doshi, 2005). Uncertainty therefore covers not only the world but also others’ information, preferences, and policies. This view underlies opponent modelling and belief-space coordination (Carmel & Markovitch, 1996; Huang et al., 2024; Goel et al., 2026); the observation–belief–action links nevertheless remain fragile for LLM agents (Sobotka et al., 2026; Jha et al., 2026). Let $b_t^i \in \Delta(IS_i)$ be agent i ’s belief conditioned on public history h_t , its own preferences v_i , and its private observations r_t^i . An observation updates it as $b_{t+1}^i = \tau_i(b_t^i, a_t^i, o_{t+1}^i)$, while an action is selected by belief-conditioned planning: $a_t^{i,*} \in \arg \max_{a \in A_i} \mathbb{E} \left[\sum_{k=t}^T \gamma^{k-t} R_i(x_k, a_k) \mid b_t^i, a_t^i = a, \pi_{-i} \right]$. We call these computations *inference*, which maintains the part of b_t^i predictive of others, and *planning*, which evaluates actions under that belief. They form a loop: inference informs an action, the action determines the next observation, and that observation informs the next inference. For this distinction to be experimentally useful, the hidden state must be both inferable and consequential: interaction must provide evidence that supports a reasonable belief about another agent, and different beliefs must affect the relative value of available actions. We therefore need decision points at which evidence, private information, and action consequences are all specified well enough to evaluate the belief and the choice separately. In the game below, we operationalize inference with an inference target and planning with counterfactual action values.

2.2 NEGOTIATION WITH PRIVATE PREFERENCES AND MIXED INCENTIVES

To make inference and planning separately observable while retaining exact supervision, we extend the commitment-based negotiation game of Benac et al. (2026) with private preferences and costly information gathering. An instance of this game is

$$\mathcal{E} = \langle \mathcal{N}, \mathbf{K}, \mathcal{G}, \Gamma, \delta, \rho, C_0, \mathcal{L}, \{\Theta_i\}_{i \in \mathcal{N}}, \mu_0, \mathbf{q}_0 \rangle. \quad (1)$$

Here $\mathcal{N}$ is the player set and $\mathbf{K} = (K_i)_{i \in \mathcal{N}}$ gives each player’s number of binary commitment coordinates. Each public goal $g \in \mathcal{G}$ has a nonempty requirement set $\Gamma_g \subseteq \{(i, k) : i \in \mathcal{N}, 0 \leq k < K_i\}$ and a scoring mode $\delta_g \in \{\text{binary}, \text{linear}\}$. The public schedule $\rho = (\rho_t)_{t=0}^{T-1}$ specifies the proposer at each of T proposal opportunities, and $\mathcal{L}$ gives the legal actions at each information set. The commitment state is $C = (c_{ik})$, with $c_{ik} \in \{0, 1\}$; initially $C_0 = 0$, and a commitment cannot be withdrawn. Each player privately observes its preference vector $v_i \in \Theta_i \subseteq \{-1, 0, +1\}^{|\mathcal{G}|}$. The joint type $v = (v_i)_{i \in \mathcal{N}}$ is drawn from the common prior $\mu_0 \in \Delta(\prod_{i \in \mathcal{N}} \Theta_i)$; entries encode AVOID, NEUTRAL, and WANT, respectively. $\mathbf{q}_0 = (q_{i,0})_{i \in \mathcal{N}}$ gives the initial investigation budgets. For any commitment state C , goal satisfaction and terminal utility are

$$S_g(C) = \begin{cases} \mathbf{1}[c_{ik} = 1, \forall (i, k) \in \Gamma_g], & \delta_g = \text{binary}, \\ \frac{1}{|\Gamma_g|} \sum_{(i,k) \in \Gamma_g} c_{ik}, & \delta_g = \text{linear}, \end{cases} \quad U_i(C; v_i) = \sum_{g \in \mathcal{G}} v_{ig} S_g(C). \quad (2)$$

At initialization, the goals, prior, legal-action rules, and proposer schedule are public, while some of the preferences are hidden; every player starts with one investigation opportunity ($q_{i,0} = 1$). Play follows ρ . At each proposal opportunity, the scheduled player may PASS, make a bilateral OFFER to a chosen partner, or, if $q_{i,t} > 0$, use INVESTIGATE(j, g) to learn v_{jg} . Passing leaves commitments unchanged. Investigation also leaves them unchanged, uses that proposal opportunity and one unit of budget, and reveals the answer only to the investigator; the query target is public. An offer specifies new commitments for the proposer, the partner, or both. The partner immediately chooses

ACCEPT or REJECT, acceptance makes all proposed additions binding and irreversible, whereas rejection changes no commitments. The response does not use another proposal opportunity. Play then proceeds to the next scheduled proposer and ends after the final opportunity, when each player receives $U_i(C_T; v_i)$. To improve its utility, a player infers partners' preferences from offers, responses, and any private investigation answers, then plans whom to approach, what to offer, and whether to accept, accounting for goals on which preferences align or conflict. Figure 2 illustrates these transitions in a three-player game. Appendix A provides the full formal specification.

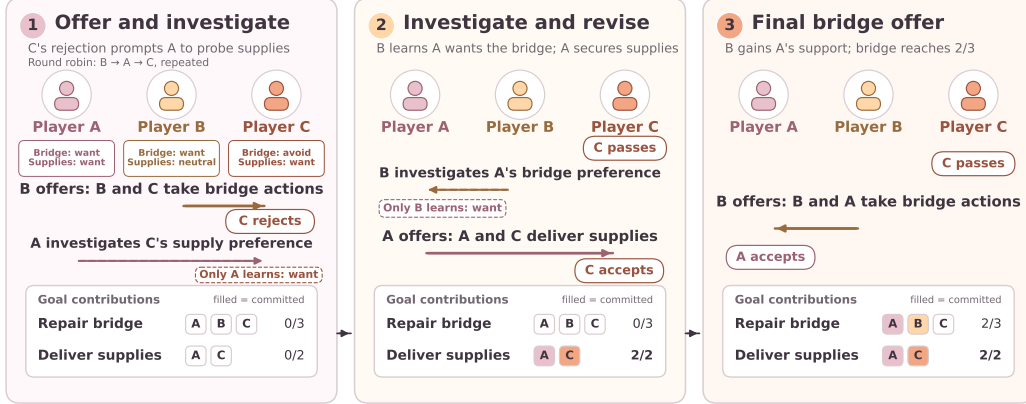


Figure 2: An illustrative three-player negotiation with private preferences and two linearly scored goals. *Repair bridge* requires distinct bridge actions from A, B, and C; *Deliver supplies* requires distinct supply actions from A and C. A filled letter indicates that player's commitment to that particular goal, and the fractions show goal progress. B first offers bridge actions to C, who rejects: accepting would leave the bridge at 2/3 and give C $-2/3$ from a goal it dislikes if play stopped then. A investigates C's supply preference and privately learns that C wants supplies. After C passes, B investigates A's bridge preference and privately learns that A wants the bridge. A then offers to deliver supplies with C, who accepts, completing that goal. C passes again before B offers bridge actions to A, who accepts, bringing bridge progress to 2/3; C does not join it. The game ends after this final offer. Proposer turns follow the fixed round-robin order B–A–C throughout.

2.3 GAME SLICES AND ORACLE SOLVER

A *game slice* is a consecutive segment of play that begins at a chosen decision and may include later decisions. For example, it can contain an offer, the partner's response, and the next proposal. The slice records which actions occurred and what each player observed following the game rule. Whereas a full game starts with zero commitments, a slice may start from a legal history prefix or an explicitly specified exogenous state, with its current commitments and remaining investigation budgets recorded. At decision s , the acting player i knows the public history h_s , its own preferences v_i , and its private answers r_s^i . We write this view as $I_s^i = (h_s, v_i, r_s^i)$, an *information set* that identifies the game states the player cannot distinguish (Kuhn, 1953). The observed sequence locates the decisions to be labeled. For each such decision, the oracle also evaluates actions that were not taken and their possible continuations; it does not give the acting player access to the partners' hidden preferences. A formal slice definition appears in Appendix B. Following the Bayesian representation of private types (Harsanyi, 1967), the oracle solves the finite tree over possible preferences and actions under the common prior μ_0 . The oracle labels are exact relative to the public prior, the specified partner policies, and the fixed continuation policy. A *policy* $\pi_i(a | I)$ assigns a probability to each legal action $a \in \mathcal{L}(I)$ given what player i can know. Let Π_i contain all such policies, including plans for later decisions after each possible observation; π_{-i} denotes the other players' policies. Holding π_{-i} fixed, a *best response* maximizes player i 's expected terminal utility over these complete plans. Starting from uniform legal actions, we update every player against the same previous policy profile:

$$\begin{aligned} \text{BR}_i(\pi_{-i}) &\in \arg \max_{\pi_i \in \Pi_i} \mathbb{E}_{\mu_0, (\pi_i, \pi_{-i})} [U_i(C_T; v_i)], \\ \pi_i^{(0)}(a | I) &= |\mathcal{L}(I)|^{-1}, \quad \pi_i^{(k+1)} = \text{BR}_i(\pi_{-i}^{(k)}). \end{aligned} \quad (3)$$

We retain a fixed profile π^* only if it passes a no-profitable-unilateral-deviation check, following the stability criterion of Nash (1950): in every initial private-information class, no player can improve its expected utility by changing its complete plan while the others keep theirs. At ACCEPT/REJECT responses, expected utility to the other players breaks ties in the responder’s own utility; remaining ties are mixed uniformly. The reference policy also turns interaction into an inference target. Let $b_s^{i,*}(v_{-i}) = \Pr_{\mu_0, \pi^*}(v_{-i} \mid I_s^i)$ be player i ’s *posterior*, its probability distribution over the others’ hidden preferences. If partner j autonomously chooses a_s^j , Bayes’ rule gives

$$b_{s+1}^{i,*}(v_{-i}) \propto b_s^{i,*}(v_{-i}) \pi_j^*(a_s^j \mid I_s^j(v_i, v_{-i})), \quad j \neq i. \quad (4)$$

An action is evidence to the extent that the reference policy makes it more likely under one candidate preference assignment than another. For planning, we instead ask what would happen if i chose each legal action a now:

$$Q_i^*(I_s^i, a) = \mathbb{E}_{\mu_0, \pi^*}[U_i(C_T; v_i) \mid I_s^i, \text{do}(a_s^i = a)]. \quad (5)$$

The forced action is followed by the reference policy for the rest of the game. Across a longer slice, each player’s posterior is updated from the observations available to that player and informs its later decisions. Appendix B details the computation and validity checks; Section 4.1 describes slice construction and sampling.

3 DIAGNOSIS

An unsuccessful interaction does not reveal whether an agent misread its partner or chose poorly despite an adequate understanding. We diagnose these two sources of error in the Qwen3-4B-Instruct-2507 model (Yang et al., 2025). At matched decision points, we ask whether the model can infer a partner’s preference from available evidence, choose a good action given a correct inference, and improve its choice when an incorrect inference is replaced by an oracle judgment.

Intervening on the inference supplied to planning. Each case elicits four responses. The model first reports a preference judgment $\hat{J}_c$ from the history and available private information. Two separate planning calls then receive the same current state, with either that judgment ($\text{Plan}_{\text{model}}$) or the oracle judgment J_c^* ($\text{Plan}_{\text{oracle}}$). We also evaluate *direct action selection* from the original state, without a supplied judgment. The paired planning calls share their tools, decoding settings, and seed. We remove the behavioral history from both calls so that planning uses the supplied inference rather than reconstructing it from the original evidence. To ensure that this removal does not make the task underspecified, we retain only cases in which the oracle judgment determines the optimal action set and removing history preserves the continuation payoff table.

Evidence and oracle labels. The suite contains 22 two-player cases spanning seven game structures. Six test use of a preference revealed by investigation; six require ruling out preferences after one observed partner choice; six retain uncertainty after one choice; and four require combining successive observations. Each case has one unknown partner–goal preference, taking values WANT, NEUTRAL, or AVOID. The oracle conditions on all available evidence under the public prior and reference policy of Section 2.3. Its label specifies the *support*, the preferences with positive posterior probability, and the *most likely preference*. We report the latter only when its posterior probability exceeds the next highest by more than 0.1; otherwise, we report UNDETERMINED. These labels describe what the evidence supports and we verify that all posteriors compatible with each oracle label yield the same optimal action set. Appendix C details these constructions.

We report joint inference correctness: a judgment is correct only when both its support and most likely preference match the oracle label. Action correctness uses the oracle’s acceptable-action set, with the utility tolerance specified in Appendix C. To measure the cost of an error and the effect of replacing a judgment, let $V_c(a) = Q_i^*(I_c, a)$ be the expected terminal utility of action a under the oracle posterior and reference continuation. We report

$$\text{Reg}_c(a) = \max_{a' \in \mathcal{L}(I_c)} V_c(a') - V_c(a), \quad \Delta_c = V_c(\hat{a}_c(J_c^*)) - V_c(\hat{a}_c(\hat{J}_c)), \quad (6)$$

where $\hat{a}_c(J)$ is the model’s planning response to judgment J . Both actions are scored under the same posterior and continuation, $\Delta_c > 0$ means that replacing the judgment improves the selected action’s value.

Table 1: Diagnosis of the unchanged Qwen3-4B-Instruct-2507 model. **Top:** correct responses out of all cases. Joint inference requires both the support and most likely preference to be correct; Direct selects an action from the original state. **Bottom:** paired intervention results, grouped by whether the model’s initial judgment was correct. One invalid $\text{Plan}_{\text{model}}$ response counts as incorrect above and is excluded below. Equal action accuracy can conceal changes in action value.

Evidence	Inference		Action	
	Joint	Direct	$\text{Plan}_{\text{model}}$	$\text{Plan}_{\text{oracle}}$
Direct feedback	6/6	2/6	3/6	3/6
Single-choice elimination	0/6	3/6	2/6	3/6
Ambiguous single choice	5/6	2/6	2/6	2/6
Multi-step update	0/4	4/4	4/4	3/4
All cases	11/22	11/22	11/22	11/22

Initial judgment	Pairs	Action changes	$\Delta > 0$	$\Delta = 0$	$\Delta < 0$	Mean Δ
Incorrect	10	9	3	6	1	0.300
Correct	11	0	0	11	0	0.000
All valid pairs	21	9	3	17	1	0.143

Table 2: A two-goal inference error and its correction. Both goals are binary and require both listed commitments. Alex wants both goals; only Blair’s Orchard preference is hidden. The two planning calls use the same model and state, with different supplied judgments. Utilities are evaluated under the oracle posterior and reference policy.

(a) Goals and preferences					
	Required commitments			Preferences	
	Goal	Alex	Blair	Alex	Blair
	Orchard	Cedar	Cedar	Want	Hidden
	Harbor	Maple	Cedar	Want	Want
History	Blair offers to add nothing and asks Alex to add Cedar; Alex accepts.				
Current decision	Alex has Cedar; Blair has none. Alex has the final proposal turn.				
Key implication	After Alex adds Cedar, any completion of Harbor also completes Orchard. A Blair that avoids Orchard could instead have secured Harbor alone.				
Model judgment	Possible preferences: Avoid only; most likely: Avoid.				
Oracle judgment	Possible preferences: Want or Neutral; most likely: Want.				
(b) Final offers and outcomes					
Action source	Alex adds	Blair adds	Goals completed	U_{Alex}	
Plan _{model}	None	Maple	None	0	
Plan _{oracle}	Maple	Cedar	Orchard and Harbor	2	

Inference and planning failures. The model recovers all six directly revealed preferences but fails all ten single-choice elimination and multi-step judgments (Table 1). Even with oracle inference, planning succeeds in only 11/22 cases, with mean regret 0.567. Correcting an incorrect judgment changes the action in nine of ten valid pairs: three pairs improve in value, six preserve it, and one worsens. Across all 21 valid pairs, mean regret falls from 0.684 to 0.541 despite unchanged aggregate action accuracy. Thus both behavioral inference and action selection exhibit failures; correcting the supplied inference can improve choices but does not reliably resolve planning errors. Appendix C provides the breakdown by sensitivity to partner preferences.

An illustrative intervention. In Table 2, Blair asks Alex for a commitment that makes completing Harbor also complete Orchard. Since Blair could instead secure Harbor alone, this choice rules out AVOID for Orchard under the reference policy. The model infers the opposite from Blair’s lack of its own commitment. Replacing that judgment changes its offer from completing neither goal to completing both, raising utility from 0 to 2. Appendix C.1 gives the derivation and recorded outputs; Appendix C.2 provides a case where correction leaves a planning error.

Together, these probes reveal failures in extracting preference information and in using sufficient information to choose an action. The intervention connects the two at the explicit judgment–action interface. These findings motivate training both capabilities alongside direct action selection; the effects of training and transfer are evaluated separately below.

4 EXPERIMENT

4.1 DATA AND TRAINING

Constructing games and histories. We build training data from the finite BENAC-I games in Section 2.2, varying goal dependencies, binary or linear satisfaction, private preferences, and public priors. From these games, we construct legal interaction histories and extract decision points together with the information available to the acting player. We also branch on investigation outcomes to obtain related decisions in which newly revealed preferences can change the appropriate action. Each history is replayed under the native game rules to verify its transitions. This construction provides targeted examples of interpreting behavior, incorporating new evidence, and choosing actions under different preference alignments.

Generating supervision. At each selected decision, the oracle in Section 2.3 conditions on the player’s observations and computes expected terminal utilities for legal actions under the reference continuation. These computations yield three training tasks: direct action selection from the observed state and history; inference of a partner’s possible and most likely preferences; and planning given an oracle inference. Action labels retain the acceptable actions rather than imposing a single answer when actions tie. We exclude failed reference computations and mask planning labels that remain ambiguous under the supplied qualitative judgment. In the paired bank, related views and histories from the same source family share a split, yielding 373 training and 124 validation cases. The reported checkpoints also include an earlier inference–planning dataset and a compact joint training set of 24 items selected using prior training performance. Appendix D records these dataset variants and their selection procedures.

Training. All trained variants start from Qwen3-4B-Instruct-2507 (Yang et al., 2025). The unchanged model provides the baseline. Self-play (SP) learns from each player’s terminal utility in complete games. Slice training instead uses oracle-scored responses: GAMESLICE-A trains direct action selection, GAMESLICE-IP trains inference and conditional planning, and GAMESLICE-JOINT combines all three tasks. These objectives update a single model without imposing an inference-to-planning pipeline at test time. Slice training uses grouped policy optimization with eight responses per item and a KL penalty toward the frozen base model. We use BF16 and an initial learning rate of 10^{-6} . Table 7 provides the remaining settings and the realized training dose for each verified checkpoint; the reported variants differ in both data and training budget.

4.2 IN-DOMAIN DIAGNOSIS AND ADVERSARIAL GAMES

The archived post-training runs use an earlier 16-case diagnostic (v6), distinct from the 22-case base-model diagnostic (v9) in Section 3. Table 3 compares checkpoints within that earlier protocol; its baseline scores are not directly comparable to Table 1. We report exact semantic inference judgments, planning regret when the correct judgment is supplied, native history-to-action regret, and the paired judgment-repair effect. Regret is lower-is-better; repair is computed only where both paired actions are valid. The three deterministic repetitions of a structure are not independent test cases.

We also test full BENAC-I games with one focal model and two frozen baseline partners. Initial worlds are partitioned into seen dependency structures with new preference assignments (ID) and structurally held-out dependencies (OOD); focal seats are rotated. We report completed episodes and focal terminal utility conditional on completion separately, because an invalid or truncated game has no terminal utility. The currently archived comparison below is an earlier 48-episode, one-sample-per-scenario-seat evaluation and does not cover the selected O and D checkpoints.

4.3 TRANSFER TO COMPLEX SCENARIO

CalBench setting. CalBench (Zou et al., 2026) studies decentralized meeting scheduling: agents see their own calendars and displacement costs, exchange messages with other participants, and decide whether to move private errands or existing meetings to agree on a common slot. We evaluate homogeneous teams, with the same checkpoint controlling all four agents. Each game has eight calendar slots and three meetings that arrive sequentially, so an early commitment can constrain a

Table 3: Historical post-training diagnosis under the earlier v6 protocol: 16 partner-generated histories, one per held-out structure. These are different cases from the v9 diagnostic in Section 3. Planning regret uses the correct inference judgment. Regret and repair are means over valid outputs; their coverage may differ. Blank cells await a verified run under this protocol. BP-99 is a historical inference/planning-only reference, not one of the four main arms. Valid planning/native/repair counts are Q0 16/15/12, D-72 13/16/9, D-148 14/16/11, and BP-99 11/14/6.

Model	Inference exact	Planning regret	Native regret	Repair Δ_{repair}
Baseline (Q0)	0%	0.844	0.933	-0.042
SP	—	—	—	—
O	—	—	—	—
D-72	0%	0.769	0.750	+0.111
D-148	0%	0.821	0.469	0.000
BP-99 (historical)	0%	0.773	0.679	0.000

Table 4: Historical fixed-partner adversarial games. Utility is conditional on normal completion; completion counts are out of 24 per split. The final common-protocol O/D comparison remains to be filled.

Model	ID complete	ID utility	OOD complete	OOD utility
Baseline (Q0)	22/24	0.288	22/24	0.068
SP-69	20/24	0.525	18/24	-0.019
O	—	—	—	—
D	—	—	—	—
BP-139 (historical)	19/24	0.254	21/24	0.333

later decision. Our frozen 24-game suite crosses four conditions (loose availability, dense calendars, blocked errands, and a replanning control), two errand-cost regimes (uniform or varied), and three scenario seeds. The replanning control can require contacting a participant in an earlier meeting who is absent from the new one. This is a small-model sequential adaptation of CalBench, not its original aggregate benchmark score.

All models use the same cases, direct messages only, temperature zero, a 4,096-token output limit, and a 32,768-token context limit. Each meeting allows four communication turns per participant and one decision retry; we disable fallback and reflection. The cases have private calendar occupancy and costs, but no future meeting or evaluator reference appears in an agent’s observation. The Qwen3-4B thinking model, MARSHAL Generalist (Yuan et al., 2026), and Social-R1-4B (Wu et al., 2026) form a separate comparison group because they use thinking-mode checkpoints; the baseline and our trained models use Qwen3-4B-Instruct-2507. We call the slice-trained variants GAMESLICE-A (history-to-action; O-99), GAMESLICE-IP (inference and conditional planning; BP-99), and GAMESLICE-JOINT (all three objectives; micro24-39). The names identify their training signals; the checkpoints also differ in exposure.

Evaluation. In our sequential suite, we count a meeting only if it remains in one common slot for all participants in the final calendars, and count a complete game only if all three meetings remain valid and blocked items are preserved. Thus the first two columns of Table 5 have denominators 72 and 24. For a completed game g , per-agent excess cost is $(C_g - C_g^*)/4$, where C_g is realized team rescheduling cost and C_g^* is the minimum feasible cost certified by a sequential replay; we average this quantity over completed games only. The reference has access to the full scenario after the fact and is not an online policy. Communication is the native per-agent count of direct messages divided by the number of meetings that agent helped schedule averaged over four agents and 24 games. Burden imbalance is the native mean absolute deviation of agents’ excess burdens from their game-level mean, averaged over the 24 games.

GAMESLICE-JOINT retains the most meetings, while GAMESLICE-IP completes the most full games. Their strengths also differ by condition: GAMESLICE-JOINT retains 17 of 18 meetings in the loose cases and 15 of 18 in the replanning cases, compared with 14 and 10 for the unchanged baseline; GAMESLICE-IP retains 10 of 18 in the blocked cases, compared with 6 for the baseline and 5 for GAMESLICE-JOINT. A matched initial scenario, with verbatim messages in Appendix E, illustrates the coordination behind the former result. In `replan_varied_sl`, GAMESLICE-JOINT

Table 5: CalBench transfer on 24 frozen sequential games. Meetings count final retained meetings; excess cost is per agent and conditional on full-game success. Communication and imbalance average native per-agent values over all games. Best value in each column is bold.

Model	Meetings $\uparrow$	Full games $\uparrow$	Cost $\downarrow$	Comm. $\downarrow$	Imbalance $\downarrow$
Baseline	37	5	0.100	5.663	0.443
Self-play	37	6	0.625	5.181	0.755
GAMESLICE-A	38	8	0.688	5.003	0.490
GAMESLICE-IP	44	10	0.075	4.760	0.432
GAMESLICE-JOINT	47	9	0.194	4.714	0.292
Qwen3-4B (thinking)	42	7	0.571	4.828	0.557
MARSHAL Generalist	37	6	1.333	4.429	0.875
Social-R1-4B	39	5	1.100	4.554	0.901

initially proposes slot 2, learns that it is blocked for one partner, checks that the proposed alternative (slot 1) works for the other, and all three participants schedule the first meeting at slot 1. When slot 0 is later proposed for the third meeting, two participants identify its conflict with the second meeting and the team converges on slot 2. All three meetings remain valid. Under the same initial calendars, the baseline and self-play teams each retain only the first meeting: in the second round a participant reports that slot 1 is blocked, yet the resolution records different slots.

5 DISCUSSION

The game that we modified makes the connection between inference and planning measurable through explicit information sets and verified action values. We want to discuss three thought.

Richer partner states. Our diagnosis concentrates uncertainty on a partner’s preferences. In less constrained interactions, an agent must also infer what its partner has observed, what it believes others know, and which behavioral rule it follows. The same refusal, for example, could express an unfavorable preference or uncertainty about information unavailable to the partner. Recent work on multi-agent theory of mind examines private knowledge, adaptive reasoning about another agent’s mental state, and hypotheses about partner strategies (YS et al., 2026; Mu et al., 2026; Cross et al., 2025). These are extensions of *inference* as defined here: each concerns a latent state needed to predict a partner’s response and plan one’s own action. Extending game slices to such states would test whether the separation between inference and planning remains useful when beliefs themselves have structure.

Strategic coverage. The appeal of game slices lies in control over which decisions become training examples. Complete-game self-play generates coherent trajectories with the learner’s current partners, but its training distribution is the set of nodes those policies actually reach. A rarely visited offer, rejection, or information-gathering decision may therefore receive little training signal even when it is strategically decisive. We instead construct and sample slices around specified interaction patterns, retaining cases for which the reference computation supplies verified inference and action-value targets. Work on subgame curricula likewise shows that changing the distribution of starting states can accelerate multi-agent learning (Chen et al., 2024). This offers a possible explanation for the efficiency of targeted slices, though our comparison does not isolate state selection from denser supervision. The complementary strengths suggest a concrete hybrid: use complete games to discover consequential failures, solve and train on slices around those decisions, then return to full interactions to test whether the learned behavior survives its wider context.

Robustness across partner policies. Our reference labels assume one selected partner policy, while self-play primarily exposes the learner to copies of its evolving policy. This matters for inference as well as action selection: an offer that is impossible under one reference policy may be ordinary for a partner with a different planning horizon, tie rule, or degree of noise. Evidence that self-play agents can learn conventions that fail with unfamiliar partners, and that diverse training partners can improve cross-partner coordination motivates testing policy variation directly (Hu et al., 2020; Strouse et al., 2021). One direction is to train against a population of partner policies and generate slice labels under each policy, including controlled deviations from the reference

solver. Belief targets could then marginalize over uncertainty about both preferences and behavioral policies, while evaluation holds out entire policy families. Such tests would distinguish strategic behaviors that transfer across partners from responses tailored to the particular oracle used here.

6 RELATED WORKS

Classical multi-agent decision making couples inference about other agents with downstream action selection, from opponent models and interactive beliefs (Carmel & Markovitch, 1996; Gmytrasiewicz & Doshi, 2005; Albrecht & Ramamoorthy, 2013) to belief-space coordination, inference from actions, and opponent-aware planning (Macindoe et al., 2012; Wu et al., 2021; Foerster et al., 2019; Huang et al., 2024). For LLM agents, explicit mental-state or behavioral representations support collaboration, adaptation, trust tracking, and planning (Li et al., 2023; Cross et al., 2025; Jha et al., 2026; Mu et al., 2026; Zhou et al., 2026; Feng et al., 2026; Goel et al., 2026; Hwang et al., 2026). However, accurate mental-state prediction need not yield useful behavior (Riemer et al., 2025; Sobotka et al., 2026).

Recent MAS benchmarks expose strategic failures under private information, mixed incentives, and imperfect communication. Commitment-based negotiation, self-negotiated contracts, and multi-party conquest model consequential agreements and coalition dynamics (Benac et al., 2026; Wyse et al., 2026; O’Neill et al., 2026); CalBench, SOTOPIA-TOM, and SNEAK make information disclosure and privacy strategically consequential (Zou et al., 2026; YS et al., 2026; Avsian & Heck, 2026). Broader evaluations cover collaboration, controlled interaction, long-horizon commerce, and communication barriers (Zhu et al., 2025; Chen et al., 2026; Li et al., 2026b; Xuan et al., 2026). Diagnostic work further finds failures in distributed information pooling, distributed computation, dynamic grounding, and belief-sensitive negotiation (Li et al., 2026a; Zhang et al., 2026b; Yao et al., 2026; Zhang et al., 2026a; Agashe et al., 2025); these complement broad failure taxonomies and accounts of long-lived peer interaction (Cemri et al., 2025; Anthropic, 2026).

Game-based training has been used to induce transferable strategic and social reasoning through self-play, multi-game reinforcement learning, and social reasoning supervision (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b; Hua et al., 2026; He et al., 2026a; Wu et al., 2026). Cross-environment cooperation and Other-Play study generalization across tasks and partners (Jha et al., 2025; Hu et al., 2020). Curriculum work selects terminal-proximal or high-learning-value subgames (West et al., 2020; Chen et al., 2024), while LLM curricula can also induce undesirable strategic abstractions (Madmoun & Lahlou, 2026). Similarity-based inference has also been proposed as a route to cooperation among foundation-model agents (Meulemans et al., 2026).

7 CONCLUSION

AI USE STATEMENT

We used generative AI tools to assist with literature discovery, reference formatting, and drafting and editing portions of this manuscript. All AI-assisted text and bibliographic metadata were reviewed by the authors. The authors take responsibility for the final content of the paper, including its claims, experimental artifacts, and citations.

ETHICS STATEMENT

This work studies LLM-agent interaction in a synthetic, simulated negotiation environment and does not involve human participants or personal data. The environment includes mixed incentives and strategic information disclosure, which can surface behaviors such as manipulation or unhelpful information sharing. We use these behaviors only for controlled evaluation and training of coordination capabilities. We document the game rules, oracle policies, and limitations so that the resulting methods are not interpreted as deployment policies for real-world autonomous agents.

REPRODUCIBILITY STATEMENT

The main text and appendix specify the game rules, information structure, oracle belief filtering, and reference planning procedure. The appendix reports the diagnostic positions, prompts, evaluated model version, decoding configuration, aggregation procedure, random seed, and provenance information used for the reported results. These details are intended to make the diagnostic suite and its measurements reproducible.

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A FORMAL GAME SPECIFICATION

This section completes the game definition in Section 2.2. The commitment representation, goal satisfaction functions, and terminal utility follow the action-level negotiation framework of Benac et al. (2026). We specify the information available to each player and the transition rules used in our private-preference setting.

Public instance and private types. The public instance specifies the players, commitment coordinates, goal requirements, scoring modes, proposer schedule, and legal-action rules in Eq. 1. A commitment coordinate belongs to a player; it is not a goal, an item transferred to another player, or an exclusive choice among alternatives. Different goals may depend on the same coordinate. Once that coordinate becomes binding, it contributes to every goal that requires it. A goal’s requirements do not indicate which players want that goal: requirements and preferences are separate parts of the instance.

The preference assignment v is sampled once and remains fixed throughout play. Each player observes its own complete vector v_i . Any publicly specified preferences are common knowledge as well. Write $F = \{(i, g, z)\}$ for these fixed facts and $\mathcal{W} = \text{supp}(\mu_0)$ for the admissible joint assignments. In instances generated from public categorical weights w_z , the prior is obtained by conditioning independent draws of the unfixed entries on the stated generation constraints:

$$\mu_0(v) = \frac{1}{Z} \mathbf{1}[v \text{ agrees with } F] \mathbf{1}[v \text{ satisfies the generation constraints}] \prod_{(i,g) \notin \text{dom}(F)} w_{v_{ig}}. \quad (7)$$

Here Z normalizes the distribution. The constraints used in these instances require each player to want at least one goal and each goal to have at least one non-neutral player. The weights and fixed facts are provided with the instance; they are not inferred from the realized world. Conditioning on these constraints can create dependence between preferences, so independent background draws do not imply independent posterior beliefs.

State and decision phases. A full environment state can be written as

$$x = (t, C, p, \mathbf{q}, v, \mathbf{r}, h), \quad (8)$$

where t indexes proposal opportunities, C is the binding commitment matrix, p is either an unresolved offer or $\emptyset$, $\mathbf{q}$ contains remaining investigation budgets, $\mathbf{r} = (r^i)_i$ contains private investigation records, and h is the public event history. This full state is available to the environment, not to every player. When $p = \emptyset$ and $t < T$, the actor is ρ_t . When an offer is pending, its recipient acts instead. The index t does not advance until that offer is accepted or rejected. Thus a proposal and its response count as two decisions but only one proposal opportunity.

Legal offers. At an opportunity of player i , an offer is $o = (j, D_i, D_j)$, where $j \neq i$ and D_i, D_j are sets of new commitment coordinates owned by the two participants. Let m denote the per-player, per-offer addition limit specified by $\mathcal{L}$. Legality requires

$$D_\ell \subseteq \{k : c_{\ell k} = 0\}, \quad |D_\ell| \leq m \quad (\ell \in \{i, j\}), \quad |D_i| + |D_j| > 0. \quad (9)$$

Any additional forbidden coordinates specified by the instance are excluded. One participant's addition set may be empty. The limit applies separately to each participant in this offer; it is not a lifetime commitment limit. Previously binding coordinates cannot be removed or submitted as new additions. The native state representation retains those coordinates when constructing the resulting commitment vectors. The game uses bilateral offers, not menus of alternative offers.

Transitions. When no offer is pending, the scheduled proposer may take one of the following actions:

- **PASS:** append the public action to h , leave C unchanged, and advance t by one.
- **OFFER**(j, D_i, D_j): append the offer to h and set $p = o$. No commitment binds yet and t is unchanged. The recipient must then choose **ACCEPT** or **REJECT**. Acceptance sets $c_{\ell k} = 1$ for $k \in D_\ell$, $\ell \in \{i, j\}$; rejection leaves C unchanged. Either response is public, clears p , and advances t .
- **INVESTIGATE**(j, g): legal only if $j \neq i$ and $q_i > 0$. The environment appends the public query (i, j, g) to h , privately appends (j, g, v_{jg}) to r^i , reduces q_i by one, and advances t . Commitments are unchanged. Other players' private records are unchanged.

Investigation is allowed even when the queried preference is already known or does not affect the current decision. Its cost is the consumed proposal opportunity and budget, not a separate utility penalty. It is unavailable during a pending response. Responses do not consume investigation budgets.

Observations and information sets. All players observe the public rules, fixed preference facts, commitments, proposer schedule, remaining budgets, offers, responses, passes, and query identities. An investigation answer is revealed only to its investigator. Player i 's view is $I_s^i = (h_s, v_i, r_s^i)$, with the public instance understood as fixed context. Two histories in the full game belong to the same information set for i when these views agree. In particular, two worlds may lead to the same public query while giving its investigator different private answers. The other players do not receive those answers.

Termination and incentives. A full game starts with $C_0 = 0$, no pending offer, no private query records, and $q_{i,0} = 1$. It ends when all T proposal opportunities have been used, including the response to an offer made at the final opportunity. Utilities are then evaluated by Eq. 2; there are no intermediate payoffs. For a binary goal, partial completion yields zero. For a linear goal, each required coordinate contributes an equal fraction of completion. An unachieved goal contributes zero even if it is wanted or avoided. Each player sums its own signed contributions across all goals. Utilities need not sum to zero, and one player’s improvement need not imply either an improvement or a loss for the others.

Starting a slice. A slice records its starting commitments, schedule position, pending offer if any, remaining budgets, and the observations available at that position. A state reached by a legal prefix retains the corresponding public history and each player’s own private observations. An explicitly exogenous initial state is instead a declared starting condition: it does not assert that a partner previously selected those commitments. Only actual observed choices provide behavioral evidence under the reference policy. A self-action specified as an intervention changes the physical state and may deliver a private answer, but is not itself evidence about a partner’s preference. The reference-policy computation and conditioning conventions are specified in Appendix B.

B GAME SLICES AND ORACLE SOLVER DETAILS

Multi-decision slices and information. Let $\mathcal{W} = \prod_{i \in \mathcal{N}} \Theta_i$ be the finite set of joint preference assignments, with prior μ_0 from Eq. 1. For a segment spanning decisions t through $u - 1$, a teacher-side record is

$$\sigma_{t:u} = (\mathcal{E}, v, h_t, (a_s, o_{s+1}, m_s)_{s=t}^{u-1}), \quad t < u, \quad (10)$$

where $v \in \mathcal{W}$ is the sampled joint preference assignment, h_t is the starting public history, a_s is the action actually taken, and $o_{s+1} = (o_{s+1}^j)_{j \in \mathcal{N}}$ records what each player observes afterward. The marker m_s distinguishes an autonomous action from an externally imposed one. The sampled world determines private investigation answers and terminal payoffs, whereas the actions are generated by the acting players or specified by an external starting context; v alone does not determine the action sequence. Decision indices count both proposals and responses, so $u = t + 1$ gives a single-decision slice.

The public history h_s determines commitments, the pending offer, the proposer schedule, and remaining investigation budgets. Player j additionally knows only v_j and its own private answers r_s^j . A slice can contain decisions by several players, each labeled from that player’s information set I_s^j ; the sampled world is never disclosed as part of this view. The observed sequence selects decision points to label, while the oracle retains all worlds in the prior’s support and all legal terminal continuations from each such point.

Finite-tree policy selection. A policy must assign the same action probabilities in any two worlds that a player cannot distinguish. We expand the public action tree for every $v \in \mathcal{W}$, retaining world-dependent private investigation answers. For a policy profile π , its terminal utility is $V_i^\pi(n, v) = U_i(C_T(n); v_i)$. At a nonterminal node n whose actor is j , the continuation values satisfy

$$V_i^\pi(n, v) = \sum_{a \in \mathcal{L}(n)} \pi_j(a \mid I^j(n, v)) V_i^\pi(n[a, v], v), \quad (11)$$

where $n[a, v]$ is the successor after action a in world v . The private answer to an investigation affects the investigator’s later information set, not the public information available to everyone else. Here $\mathcal{L}(n)$ denotes the legal actions at node n .

In Eq. 3, BR_i is a best response over i ’s *complete contingent policy*: it can revise later decisions after every possible observation, not merely the next action. We compute it by backward induction while keeping the other players’ policies fixed. Since the plan can depend on i ’s initial private information, the maximization separates across root information classes with positive prior probability. At node n in world v , define the counterfactual reach weight

$$\lambda_{-i}(n, v) = \mu_0(v) \prod_{\substack{r \prec n: j_r \neq i \\ r \text{ autonomous}}} \pi_{j_r}(a_r \mid I_r^{j_r}(v)), \quad (12)$$

where $r \prec n$ ranges over earlier decisions on the path to n and j_r is the player acting at decision r . Conditioned on i 's information set, these weights determine the expected value of each candidate action. The product omits i 's own past choices, so an alternative plan remains evaluable even when it differs from i 's current plan; imposed moves contribute no behavioral likelihood. At a zero-reach information set, we use the root prior restricted to i 's own type and private investigation answers. We first maximize expected own terminal utility. Only at an ACCEPT/REJECT response do own-utility ties favor the action with greater expected total utility to others. Residual ties receive equal probability.

The updates for all players read the same previous profile and are applied simultaneously. A fixed profile is accepted only if, for every player and root private-information class,

$$\max_{\pi_i \in \Pi_i} \mathbb{E}_{(\tilde{\pi}_i, \pi_{-i}^*)} [U_i(C_T; v_i) \mid I_{\text{root}}^i] - \mathbb{E}_{\pi^*} [U_i(C_T; v_i) \mid I_{\text{root}}^i] \leq \varepsilon, \quad (13)$$

where Π_i contains legal information-respecting contingent policies and ε is the numerical tolerance. The check evaluates a full unilateral deviation, rather than only a one-step alternative. We discard cycles, timeouts, and profiles that fail the check. The resulting π^* is a selected reference profile, with no claim of uniqueness or of describing arbitrary human or LLM partners.

Belief filtering and action values. Let $b_s^{i,*}(v_{-i})$ be the joint posterior over all other players' types at one decision in the slice. If player $j \neq i$ autonomously takes public action a_s^j , define $\ell_s^*(v_{-i}) = \pi_j^*(a_s^j \mid I_s^j(v_i, v_{-i}))$. The candidate world determines what j privately knows, including answers to its own investigations. The exact update is

$$b_{s+1}^{i,*}(v_{-i}) = \frac{b_s^{i,*}(v_{-i}) \ell_s^*(v_{-i})}{\sum_{\tilde{v}_{-i}} b_s^{i,*}(\tilde{v}_{-i}) \ell_s^*(\tilde{v}_{-i})}. \quad (14)$$

An action by i or an imposed move changes the game state but contributes no likelihood about another player's type. A privately received investigation answer instead removes worlds inconsistent with that answer. We exclude observed actions whose total likelihood is zero. For every legal action at a selected decision, we force that action once and then follow the fixed reference profile:

$$Q_i^*(I_s^i, a) = \sum_{v_{-i}} b_s^{i,*}(v_{-i}) V_i^{\pi^*}(n_s[a, (v_i, v_{-i})], (v_i, v_{-i})). \quad (15)$$

For a queried partner-goal preference v_{jg} , the inference target is the positive-support set of $p_{jg}(z) = \sum_{v_{-i}: v_{jg}=z} b_s^{i,*}(v_{-i})$. We report the mode as the most likely preference only when its posterior probability exceeds the runner-up by more than 0.1; otherwise, we report UNDETERMINED. Planning actions are evaluated using Q_i^* , including ties among optimal actions. The student sees neither the exact posterior nor action values. We use the same selected policy throughout a slice, while each player's posterior incorporates only its own observations. The stronger guarantee that a qualitative judgment alone determines the best action is certified separately for the selected partner-history diagnostic in Appendix C.

C DIAGNOSTIC PROTOCOL AND DETAILED RESULTS

Cases and evidence. The main diagnostic uses the frozen v9 suite of 22 two-player cases spanning seven distinct game geometries. Its 12 basic cases comprise six direct feedback, three single-choice elimination, and three ambiguous single-choice cases, with two or three binary goals per basic case. Ten challenge cases comprise three additional elimination cases, three additional ambiguous cases, and four multi-step updates. Cases sharing a geometry are related probes rather than independent draws from a task population. The structures were reused during diagnostic development; this is not an untouched transfer benchmark. The v9 selection procedure uses structural and oracle criteria, not model success on the selected cases.

Initial commitments are declared starting conditions, not fictitious partner choices. Direct feedback comes from a legal focal-player investigation with a truthful private answer. That query is an intervention and need not be optimal; it does not supply behavioral evidence about the partner. Other cases use observed choices with positive probability under the specified partner policy. Labels condition on public facts, own preferences, private observations, and the relevant behavioral evidence.

Judgments and planning sufficiency. Each case queries exactly one unknown partner–goal preference, with candidate values WANT, NEUTRAL, and AVOID. The oracle judgment contains its positive-probability support and the most likely preference. We report the latter only when its posterior probability exceeds the runner-up by more than 0.1; otherwise, we report UNDETERMINED. This label does not imply equal posterior probabilities. We report joint correctness, requiring both components to match the oracle label. All labels are relative to the public prior and specified reference policy.

Planning uses the standard history-free training renderer, including tables of goal requirements and preferences and qualitative supplied beliefs. Behavioral history and numerical posterior probabilities are omitted. For each oracle judgment, exact rational polytope-vertex checks verify that all compatible posteriors induce the same optimal-action set. The admission checks also verify the scorer’s tolerance-based acceptable-action set and recompute continuation payoffs from the current state without the removed history. Every retained case passes these checks. Correctness follows the certified acceptable-action set; regret uses the exact expected-utility gap. The scorer allows an own-utility gap of at most 0.1 from the best action. At offer responses, it additionally allows a gap of at most 0.1 in the other players’ total utility relative to actions tied in own utility. The oracle policy itself uses exact utility maximization and the response tie rule; these scoring tolerances do not alter that policy.

For each possible partner preference, the qualification procedure computes the acceptable-action set under the same per-world payoff table. A case is repair-sensitive if at least one such set is disjoint from the set accepted under the oracle judgment. This is a structural criterion involving an alternative preference, not a guarantee that the model’s actual judgment error requires a different action. The remaining cases are labeled action controls in the archive. This label denotes the complement of the repair-sensitive group, not a negative control for the model’s response to its input. There are nine repair-sensitive cases and thirteen controls. All four multi-step cases are controls and each admits an acceptable action shared across all three preferences; they therefore do not require a belief-dependent change of action.

Requests and execution. Every case has four calls: inference, direct action selection, $\text{Plan}_{\text{oracle}}$, and $\text{Plan}_{\text{model}}$. The archive denotes the first two by B and O , respectively. The two planning requests share the current state, tools, decoding settings, and request seed; only the supplied judgment differs. Inference requests require a brief explanation followed by one belief-submission tool call, with explicit set semantics. Action calls use the native action tools. The intervention changes an explicit input between fresh calls rather than manipulating an internal model representation.

The reported run evaluates Qwen3-4B-Instruct-2507 once per case, at temperature zero, top- $p = 1$, top- $k = -1$, and a 4,096-token completion limit. The service uses vLLM 0.28.0, BF16, one H100 GPU, batch-invariant mode, prefix caching, and eager execution; chunked prefill and cascade attention are disabled. Runner concurrency is four and the run seed is 42. All 88 calls returned, with no truncation or infrastructure failure. One $\text{Plan}_{\text{model}}$ call has a format failure. It counts as incorrect for accuracy and is excluded from paired utility comparisons, leaving 21 valid pairs. No repeated-run uncertainty estimate is claimed.

Aggregation and intervention accounting. The repair-sensitive and control panels are reported separately. Overall counts are descriptive: joint inference correctness is 11/22; direct action selection, $\text{Plan}_{\text{oracle}}$, and $\text{Plan}_{\text{model}}$ are each correct in 11/22. Basic-case joint inference and $\text{Plan}_{\text{oracle}}$ correctness are 9/12 and 7/12, versus 2/10 and 4/10 on challenge cases.

Across 21 valid pairs, three replacements increase utility, seventeen leave it unchanged, and one reduces it. Among the ten pairs with a changed judgment, these counts are three, six, and one, respectively. There are nine action changes, including five that preserve utility. All eleven pairs with unchanged judgments reproduce the same action. Mean paired regret falls from 0.684 to 0.541, a value gain of 0.143. The all-case $\text{Plan}_{\text{oracle}}$ mean regret is 0.567 and uses a different denominator. For each pair, the value gain equals model-judgment regret minus oracle-judgment regret under the same scoring posterior. These effects can have either sign and are not an additive decomposition of the direct action-selection error.

The repair-sensitive panel has one positive and seven zero gains among its eight valid pairs. The control panel has two positive, ten zero, and one negative gain among thirteen pairs. In the single-

Table 6: Diagnostic subgroups defined by the oracle action sets. Entries are correct responses out of all cases; invalid outputs count as incorrect. The groups partition the same 22 cases as Table 1.

Case group	Inference		Action	
	Joint	Direct	Plan _{model}	Plan _{oracle}
Repair-sensitive	5/9	3/9	3/9	3/9
Action control	6/13	8/13	8/13	8/13

choice elimination layer, mean repair gain is 0.667 across six cases, but two of its three positive gains occur in controls. This distinction prevents improvements in controls from being presented as evidence that the task requires preference-dependent action selection. Of the eight valid repair-sensitive pairs, five already have the correct judgment. Among the three with an incorrect judgment, two actions change; one gains 1 in utility and two have zero gain. The mean gain is therefore 0.333 over corrected judgments and 0.125 over all valid pairs in this group. In the control group, seven judgments are incorrect; all seven actions change, with two gains, four ties, and one loss. The corresponding means are 0.286 and 0.154 over corrected judgments and all thirteen valid pairs, respectively.

C.1 TWO-GOAL EXAMPLE: RULING OUT AN AVOIDED GOAL

Table 2 uses the archived case `layer-19646c73e0c97ae729`. Starting with no commitments, Blair acts first and Alex acts last; neither has an investigation use remaining. The prior over Blair’s Orchard preference is uniform. Recomputing the reference policy gives likelihoods $(1/6, 1/9, 0)$ for the observed offer under WANT, NEUTRAL, and AVOID. Alex accepts with probability one, so the posterior is $(0.6, 0.4, 0)$ and the oracle judgment is Want or Neutral, with Want most likely. Excluding Avoid has the simple payoff argument given in the main text; the unequal weights on the two remaining preferences follow from the reference policy’s uniform mixing over tied actions.

The model instead states, “Blair’s behavior (not adding Cedar) suggests they do not want Orchard,” and submits only avoid. Its planning call requests `self.commitments=[]` and `partner.commitments=[Maple]`, completing no goal regardless of Blair’s response. With the oracle judgment, the call requests `self.commitments=[Maple]` and `partner.commitments=[Cedar]`. Blair accepts, completing both goals and giving Alex 2. The archived action indices are 2 and 4, respectively, with regrets 2 and 0 and intervention gain 2. The latter offer is also acceptable if Blair avoids Orchard: its own utility then ties rejection at 0, and the response tie rule favors Alex’s gain. This explains the case’s action-control classification and limits the conclusion to the observed benefit of correcting the model’s supplied judgment.

C.2 A POLICY-DEPENDENT EXAMPLE OF PARTIAL CORRECTION

This additional example is archived as `interaction-0524ffa3f232ff556a`. Its two proposal opportunities are Blair followed by Alex, starting with no binding commitments. Each offer can add at most one commitment per player, and commitments cannot be removed. All four goals are binary, with requirements and preferences as follows. Only Blair’s preference for Orchard is hidden; its prior is uniform over WANT, NEUTRAL, and AVOID.

Goal	Alex must add	Blair must add	Alex’s preference	Blair’s preference
Orchard	Maple	Maple	Want	Hidden
Harbor	Cedar	Maple	Want	Want
Library	Maple	Maple	Neutral	Avoid
Garden	Cedar	Cedar	Avoid	Avoid

Observed history and oracle inference. Blair proposes no commitment for itself and Cedar for Alex; Alex accepts. To evaluate this choice as evidence, the oracle compares its continuation value with the other actions Blair could have taken. Under the selected reference policy, Alex’s subsequent proposal adds Maple for both players. If Blair wants Orchard, accepting completes Orchard, Harbor, and Library, giving Blair $1 + 1 - 1 = 1$. If it is neutral, accepting gives $0 + 1 - 1 = 0$; rejection also gives 0, so the response tie rule favors acceptance for Alex’s benefit. If Blair avoids Orchard, accepting gives $-1 + 1 - 1 = -1$, so it rejects and receives 0.

Blair has an alternative first offer: add Maple for itself and Cedar for Alex. Alex accepts, immediately completing Harbor. On Alex’s last turn, adding its Maple would also complete Orchard and Library. Blair accepts this addition if it wants Orchard and otherwise rejects it. Thus the alternative secures Blair terminal utility 1 under every preference. The observed offer is optimal only under WANT, where all thirteen legal first actions tie and the reference policy mixes uniformly. Its likelihoods are therefore

$$\begin{aligned}\Pr(\text{observed offer} \mid \text{WANT}) &= \frac{1}{13}, \\ \Pr(\text{observed offer} \mid \text{NEUTRAL}) &= \Pr(\text{observed offer} \mid \text{AVOID}) = 0.\end{aligned}\tag{16}$$

Alex accepts the observed offer with probability one. Bayes’ rule maps the prior $(1/3, 1/3, 1/3)$ to posterior $(1, 0, 0)$, yielding support $\{\text{WANT}\}$ and most likely preference WANT. This inference follows from the stated reference policy and continuation, rather than from a general assumption that any offer reveals a unique preference.

Recorded model responses and their consequences. The model’s inference response instead states, “There is no evidence suggesting Blair wants or avoids Orchard,” and submits all three values with the most likely preference reported as UNDETERMINED. With that judgment, its planning tool call requests `self_commitments=[Maple]`, `partner_commitments=[Cedar]`. Accepting would complete Garden, giving Blair -1 , so Blair rejects; Alex’s terminal utility is 0. With the oracle judgment, the tool call changes to `self_commitments=[], partner_commitments=[Maple]`. Blair accepts because Harbor gives it 1; Alex also receives 1. The optimal call would add Maple for both players, completing Orchard, Harbor, and Library. Its utilities are $1 + 1 + 0 = 2$ for Alex and $1 + 1 - 1 = 1$ for Blair, so Blair accepts. The three oracle action values are therefore 0, 1, and 2, respectively. The two recorded planning calls have regrets 2 and 1, giving intervention gain $\Delta_c = 1$. These actions correspond to indices 4, 2, and 5 in the archived legal-action list. Recomputing the reference policy reproduces the archived policy hash, posterior, and action values.

Provenance. The reported archive is `diagnostic_v9_results_20260926_run_q0-882149`. Its frozen manifest and checkpoint hashes are preserved in the run protocol. The machine-readable manuscript results and source inventory identify the archive members, per-case scores, and the worked example used in the main text. Older v6 diagnostic results are not pooled with this run.

D TRAINING DATA AND CONFIGURATION

Data construction and splitting. The generators construct game states and legal histories, replay observations, and solve the resulting decisions without using an LLM to supply labels. Initial commitments specify the starting state; voluntary partner actions provide behavioral evidence under the reference policy. Investigation branches expose truthful preference feedback and the subsequent legal decisions. The paired bank renders each canonical decision as direct action selection, inference, and conditional planning. Its split groups source families, including all views of a decision and related histories. This is a split within the constructed bank: some families from earlier development validation sets were reassigned to training, so it does not establish separation from every historical evaluation suite.

Planning labels and dataset versions. For the paired bank, planning prompts supply the current state and a qualitative oracle judgment while omitting behavioral history. The label audit distinguishes actions that are acceptable, unacceptable, or ambiguous given that judgment; ambiguous labels receive no task supervision. Of the 373 training cases, 314 support planning training, 29 have no negative planning label and are reserved for evaluation, and 30 are quarantined. The corresponding validation counts are 106, 4, and 14. The historical GAMESLICE-IP checkpoint uses an earlier dataset with 294 inference and 300 planning items. These earlier planning prompts use a mixture of interfaces, as do the planning items retained for compact joint training; they should not be identified with the uniform intervention interface in Section 3.

Selecting the compact joint dataset. The 24-item joint set contains eight examples per objective, selected from training data using archived learning histories. For direct action selection and infer-

ence, the main selection rule requires at least four prior exposures, two groups with nonzero task advantages, mean accuracy of at least 50% over the final two exposures, and an improvement of at least 12.5 percentage points over the first two. One inference item has only three prior exposures and is retained as a documented exception. The eight planning items come from an earlier compact selection. Original prompts and teachers are preserved. Neither CalBench results nor validation performance enters item selection. The resulting set is frozen before training: two deterministic halves alternate across 40 updates, giving each item 20 groups of eight responses. Because selection uses prior learning performance, this run measures training on a targeted subset, rather than an isolated change in objective or exposure count.

Table 7: Data and training settings. Shared optimizer and decoding settings are verified in the archived configurations for the selected GAMESLICE-A, GAMESLICE-IP, and GAMESLICE-JOINT checkpoints. Recipe-specific rows distinguish nominal budgets from consumed response tokens. A dash denotes a quantity not yet verified for the selected checkpoint.

Parameter	Setting
Shared slice-training configuration	
Initial model / KL reference	Qwen3-4B-Instruct-2507 / frozen initial model
Precision	BF16; gradient accumulation and reduction in FP32
Optimizer	Adam, Megatron backend
Initial learning rate	10^{-6}
Adam coefficients; weight decay	(0.9, 0.95); 0
Warmup	0 steps
Policy updates per rollout batch	1 epoch
Policy-ratio clipping	0.2
KL loss coefficient	0.01
Loss aggregation	Mean over tokens within each response, then over responses
Sequence / prompt limit	4096 / 3072 tokens
Maximum generated response	1024 tokens per call
Sampling	Temperature 1; top- $p = 1$; top- k disabled
Repetition penalty	1
Responses per question group	8
Invalid or truncated output	Independent negative advantage, coefficient 0.2
Random seed	42
Training hardware layout	One node, two GPUs; tensor parallelism 2, pipeline parallelism 1
Microbatch size	1 per device
Memory settings	Sequence parallelism; full activation recomputation
Generation backend	vLLM; tensor parallelism 1 per replica
Validation / checkpoint interval	Every 10 updates
Datasets	
Paired bank	497 canonical cases: 373 training, 124 validation; 175 parent packages
Paired views	One direct-action, inference, and planning view per case; planning eligibility audited separately
Planning-eligible cases	314 training; 106 validation
Historical inference-planning set	594 training items: 294 inference, 300 planning; 495 validation items
Compact joint set	24 training items: 8 direct-action, 8 inference, 8 planning
Selected checkpoints and realized training dose	
GAMESLICE-A	O-99; 100 updates on paired-bank direct-action items
Action-only budget / consumed	6,553,600 / 6,554,714 response tokens
Action-only collection target	65,536 response tokens per update; complete groups at the boundary
GAMESLICE-IP	BP-99; 100 updates on the historical inference-planning set
Inference-planning objective weights	1/2 inference, 1/2 planning
Inference-planning budget / consumed	6,553,600 / 2,516,803 response tokens at the selected checkpoint
Inference-planning learning rate	Configured constant rate of 10^{-6}

Continued on next page

Table 7 (continued)

Parameter	Setting
GAMESLICE-JOINT	micro24-39; 40 updates on the compact joint set
Joint batch composition	4 groups per objective; $12 \times 8 = 96$ responses per update
Joint objective weights	1/3 per objective
Joint exposure per item	20 groups; 160 responses
Joint token cap / consumed	3,932,160 / 1,782,494 response tokens
Action-only / joint learning rate	Token-based cosine decay from 10^{-6} toward 10^{-7} , parameterized by each run’s token budget
SP checkpoint	SP-41; complete-game self-play with own terminal utility
SP-41 cumulative training dose	–

The response-token count measures generated tokens entering training, including retries; it is distinct from the number of unique examples or updates. The compact run terminates at its update limit before reaching its token cap. Its count follows the completed checkpoint lineage and excludes discarded work after an interrupted save. The selected SP-41 evaluation artifact identifies the model export, but its cumulative training dose remains to be linked to the training archive. Earlier SP configurations are therefore not substituted for that checkpoint. The O and D labels in the historical diagnosis tables denote action-only and joint training, respectively; those earlier D checkpoints are distinct from micro24-39.

E CALBENCH ROLLOUT EXCERPTS

The following excerpts come from the frozen `replan_varied_s1` scenario. Q0 (run 868929), self-play SP-41 (run 874283), and GAMESLICE-JOINT (run 879033) received identical initial calendars. Quoted text is verbatim `dm` content from each run’s `games/replan_varied_s1/trace.json`; intervening messages are omitted. Slot tuples below come from the environment’s round resolutions, ordered by the meeting’s participants, rather than from model explanations.

Negotiating the first meeting. GAMESLICE-JOINT’s agent 0 first proposed slot 2. Agent 1 replied:

“Hi, slot 2 is blocked for me, so I can’t use it. Slot 1 is free for me and would be a good alternative. Would that work for you?”

Agent 0 then asked agent 2:

“Hi, Agent 1 suggested slot 1. Is that free for you? I’m open to slot 1 if it works for both of you.”

Agent 2 confirmed slot 1, and the three agents’ scheduling actions all chose slot 1; the resolution tuple was (1, 1, 1).

A disagreement in the other runs. Q0 and SP-41 both placed the first meeting at slot 0. During negotiation for the second meeting, Q0’s agent 3 reported to agent 1:

“Slot 1 is blocked for me, so I can’t use it. How about slot 2? It’s free and doesn’t conflict with my schedule. Would that work for you?”

Q0’s agent 1 later responded:

“I see slot 1 is blocked for you, but it’s free for me. I don’t have any other free slots available. Could you please confirm if you’re able to adjust your schedule to slot 1? I’m happy to help coordinate if needed, but we need a shared free slot.”

SP-41’s agent 1 made the corresponding request to agent 3:

“I see you’re suggesting slot 2, but it’s blocked for me and cannot be used. Slot 1 is free and available for me without any conflict. Would you be able to join us there? If not, let me know what alternative works best for you.”

In both runs, the second meeting’s resolution tuple was (1, 2, 2), so no common slot was established. Q0’s agent 1 actually returned an empty action list at this decision; its resolution entry should not be read as an intentional slot-1 commitment.

Preserving a shared commitment. In the GAMESLICE-JOINT run, the second meeting was placed at slot 0 by all three participants. When agent 0 proposed slot 0 for the third meeting, agent 2 replied:

“Slot 0 is taken by another meeting. I suggest slot 2 instead, which is free and doesn’t conflict with any of my commitments. Would that work for you?”

Agent 3 also identified the prior meeting at slot 0. The final resolution tuple was (2, 2, 2), and all three meetings remained valid. This trajectory shows coordination around an existing commitment; it contains no action that reschedules an earlier meeting.